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Paleomagnetism of the Shakotan Peninsula, West Hokkaido, Japan

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A paleomagnetic study was made for Miocene-Pliocene volcanic and sedimentary rocks from the Shakotan Peninsula, West Hokkaido, Japan (43.2°N, 140.5°E). Out of 20 rock units sampled, 18 sites were successful to reveal stable remanence but three sites were rejected because of their anomalous directions. Although most of the sites showed in situ directions coinciding with the direction of the present earth's field (PEF), we did not adopt the usual rejection criterion; the NRM in which the in situ direction is close to the PEF should be automatically rejected. Our opinion is that as long as the original stable remanence is detected there is no need to, rather better not to, reject such NRM directions. To support this idea we compared decay curves of remanence in AF demagnetization among NRM, IRM, and ARM. The total mean paleomagnetic direction (I=55.6°, D=14.7°, α_{95}=9.5°) is slightly eastward and gives a paleomagnetic pole (77.9°N, 249.5°E, α_{95}=12.1°) which might suggest a clockwise block rotation of the Shakotan Peninsula. However, we concluded that this pole for middle Miocene to Pliocene is not significantly apart from the present geographical pole on 95% confidence level. On the same confidence level, the pole position agrees with some of the contemporary poles from Northeast Japan implying the same block of West Hokkaido and Northeast Japan. This pole position also indicates that the Shakotan Peninsula did not suffered much tectonic deformation from the collision of East Hokkaido and the Kuril arc to the middle part of Hokkaido which is considered to have occurred after the late Miocene (KIMURA and TAMAKI, 1985). The northerly paleodirection revealed by the lowest sedimentary rock (~15 Ma) might be an indication of the timing of rotation of Northeast Japan earlier than Southwest Japan.

1. Introduction

Since the pioneering work by KAWAI *et al.* (1961) the island of Japan has been studied by paleomagnetism as summarized by SASAJIMA (1981). Especially after 1980, Miocene rocks have been intensively studied and the conclusion that Southwest Japan and Northeast Japan had gone through large block rotations in opposite sense, clockwise and counterclockwise respectively, with rotation angles of around 50° for both blocks was reached (OTOFUJI and MATSUDA, 1983, 1987; OTOFUJI *et al.*, 1985a, 1985b; TOSHA and HAMANO, 1988). The rotations of the Japan Islands revealed by paleomagnetism are generally interpreted in relation to the opening of the Japan Sea which is a typical back-arc basin (HILDE and WAGEMAN, 1973; UYEDA and MIYASHIRO, 1974). The timing of the rotation is also well constrained by paleomagnetic data; Southwest Japan rotated within only 1 My of around 15 Ma (OTOFUJI and MATSUDA, 1984; HAYASHIDA

and ITO, 1984; HAYASHIDA, 1986) while Northeast Japan rotated during the period 22 to 15 Ma (OTOFUJI *et al.*, 1985a; TOSHA and HAMANO, 1988). Nevertheless, it is worthwhile to study paleomagnetism over a wide area of the Japan Island to know the extent and coherency of the rotation. Moreover, there still remains a controversy on the formation age of the Japan Sea (KONO, 1987). There was also an argument by MOREAU *et al.* (1987) on the suspicious primary origin of NRM of all the formations in Northeast Japan older than 40 Ma, which had been studied by paleomagnetism. They concluded a widespread remagnetization of pre-Oligocene rocks and suggested a rather small rotation angle of 20–30° for Northeast Japan. They even proposed that a critical review of paleomagnetism of Southwest Japan would be necessary, with emphasis on the verification of the primary origin of the NRM.

Only few paleomagnetic studies have been made in Hokkaido in order to clarify its tectonic history. Paleomagnetism of Hokkaido is important not only in relation to the opening of the Japan Sea but also to clarify the collision process of East Hokkaido to the middle part of Hokkaido which is considered to have occurred as recently as after the late Miocene (KIMURA and TAMAKI, 1985). The purpose of this study is to present a paleomagnetic result from Miocene to Pliocene rocks in the Shakotan Peninsula, West Hokkaido.

2. Sampling Localities and General Geology

The Shakotan Peninsula, West Hokkaido, locates at the northern extension of Northeast Japan and is comprised of successive sequences of volcanic and sedimentary rocks most of which were formed during Neogene (YAMAGISHI, 1981; MMAJ, 1985). The volcanic sequence is a continuation of the so-called Green tuff belt which covers Northeast Japan, this region is considered to be a part of Northeast Japan. There are mainly five formations; Pliocene Yobetsu Formation, Late Miocene Toyohama Formation, Middle Miocene Furuugawa Formation, Early Miocene Furubira, and Kayanuma Formations (Figs. 1 and 2). Classification of these formations is mainly based on ages of fossil flora found in sedimentary rocks. Thirteen volcanic units and seven sedimentary rocks were sampled, covering from the middle Miocene to Pliocene. Four volcanic rocks have multiple age data by the K-Ar method from the same unit at different sites and three of them were also dated by the fission track method (MMAJ, 1985). These multiple age data agree well with each other except for those of rhyolite from the Kayanuma Formation (ST12 and 13). Dates of this rhyolite range from 16.8 to 37.1 Ma, but the most reliable data are around 35 Ma according to MMAJ (1985), so the age of the Kayanuma Formation covers back to Oligocene.

Samples were collected with a portable rock drill. The orientation of the samples was measured with a Brunton compass which reads the magnetic north direction and sun direction when it is possible. The problem in the paleomagnetism of volcanic rocks is the difficulty of a bedding correction when tilting occurred at some time since their formation. Tilting of volcanic rocks such as the welded tuff which have an eutaxitic structure can be corrected based on the structural surface (OTOFUJI and MATSUDA, 1983) although the error of the correction is usually larger compared to that of sedimentary rock because the structural surface does not always coincide with the paleohorizontal surface (OTOFUJI *et al.*, 1985a). Dikes are also well controlled paleomagnetic targets in terms of the tilting correction problem because they usually intrude vertically to

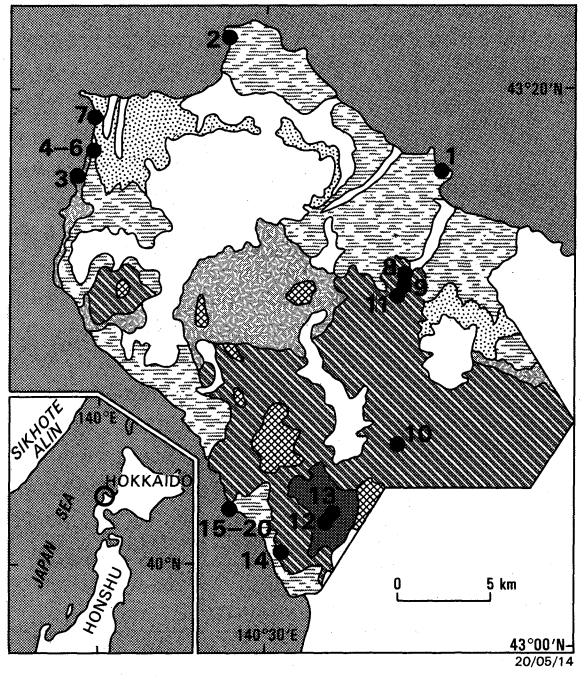


Fig. 1. Sampling locations shown on a simplified geological map of the Shakotan Peninsula. Acronym, ST, was omitted for all sites. For patterns of each geological formations, refer to Fig. 2 (redrawn after MMAJ (1985)).

the paleo-horizontal surface (TSUNAKAWA *et al.*, 1985; TOSHA and HAMANO, 1986).

In this study, we could find the strike and dip for tilting correction from most of the volcanic rocks. A dolerite body (ST03) had a definite structural surface running almost parallel to the bedding surface of the underling host sedimentary rocks. Three basalt units (ST08, 09, and 11) were considered to be lava flows because they had flow structures. The surface of the flow structure was not uniform from place to place, so the tilting correction of the NRM directions was made using the general attitude of the bedding of the host sedimentary rocks. Six pyroxene andesite dikes (ST15–20) intrude almost vertically, so we did not make any tilting correction to the NRM directions of these dikes. There remained ambiguity to the necessity of tilting correction for the amphibole andesite (ST02) and rhyolite (ST12, 13) which had no definite structural plane. However, we think the error in NRM direction caused by tilting is small, of less than 20°, for the former (ST02) because the Toyohama Formation around the sampling site is in general gently dipping or almost horizontal.

3. Paleomagnetic Study

3.1 Measurements procedure

The core samples were cut to specimens of conventional size (25 mm in diameter and 22 mm in length). Most measurements were made by a 2G Enterprise cryogenic magnetometer 740-R, and some of volcanic rock samples were measured by either the

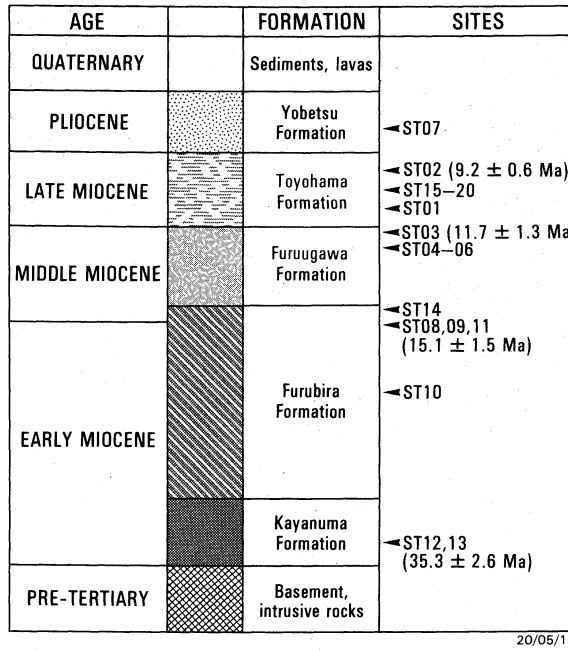


Fig. 2. Simplified stratigraphy of the Shakotan Peninsula. Stratigraphic positions of the sampling sites are roughly shown by arrows. Four volcanic units are dated by K-Ar method and three of them also have fission track datings (redrawn after MMAJ (1985)).

Schonstedt spinner magnetometer SSM-1A or a homemade spinner magnetometer of ring-core fluxgate sensor type. Usually six to eight specimens from each site were subjected to progressive demagnetization by an alternating field (AF) of up to 80 mT. Four to five more specimens were thermally demagnetized for all sedimentary rocks in progressive steps of temperature up to 500–700° C. One to two specimens were thermally treated for volcanic rock site except for some of the andesite dikes and basalt lava flows which we omitted from the thermal demagnetization because of their very stable remanences. Another exception for the demagnetization method of volcanic rocks was for hornblende andesite samples (ST02). They have an original reversed component superposed by large viscous remanent magnetization (VRM). VRMs were easily removed by AF demagnetization under 10 mT but the original remanence has such high coercivity that only thermal demagnetization could reveal it.

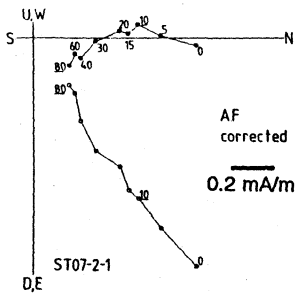
All the results of AF and thermal demagnetizations were analyzed on Zijderveld diagrams (ZIJDERVELD, 1967). Figure 3 shows one typical example for each AF and thermal demagnetizations for all sites. Only the results of AF demagnetization are shown for two sites of volcanic rocks in which thermal demagnetization was omitted. Figure 3 includes unsuccessful results where no stable remanence was determined.

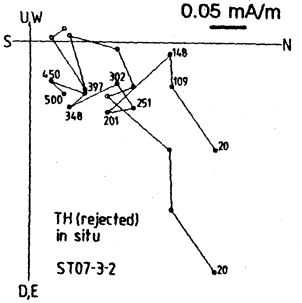
3.2 Primary origin of NRM close to the present field direction

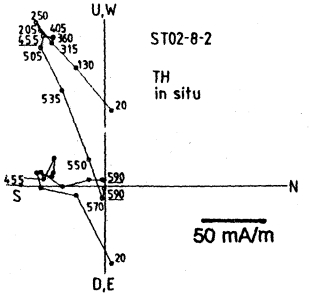
The NRM direction in the geographical coordinate, close to the direction of the present earth's field (PEF), is usually considered to be suspicious because it might

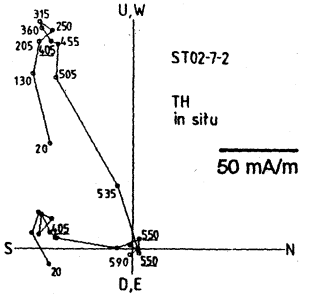
represent a VRM direction acquired during the recent age of the Brunhes normal chron. A typical criterion for the rejection of such data is whether the 95% confidence circle (α_{95}; FISHER, 1953) of the mean direction contains PEF or the dipole field direction inside of it. However, we do not think that the agreement of an in situ direction with the PEF direction is a sufficient condition to reject the site mean direction out of hand. Such a rejection criterion has the risk to reject a primary remanence which is by chance close to the PEF direction. When we encounter such suspicious data, the first thing we should do is to perform a field test such as the fold test or a baked contact test (MCELHINNY, 1973). GIDDINGS (1976) reported a NRM direction which almost coincides with the PEF direction from a group of Precambrian basic dikes in Australia where the fold test is difficult to perform because any tectonic activity such as folding had ceased since the formation of the region. In his study, the direction was also retained in the baked contact zone in a dike of another group intruded by one of the former. Even if any of these field tests is impossible to apply, we think we can judge to some extent whether the direction should be rejected or not by the stability of NRM. We think that an NRM which shows stability to both AF and thermal demagnetizations is not of VRM origin. VRM is usually AF demagnetized in fields of 20–30 mT, known from the experiment on oceanic basalt (IRVING, 1968) and stony meteorite (NAGATA, 1981). However, a surprisingly stable VRM with high coercivities of almost 100 mT has been reported for hematitebearing sedimentary rocks (BIQUAND, 1979) and artificial samples containing fine-grained hematite (DUNLOP and STIRLING, 1977). In spite of these cautionary examples, we can discount the possibility of the NRM directions in this study to be of VRM origin, because, in contrast to AF demagnetization, VRM is usually unstable to thermal demagnetization with typical unblocking temperatures of less than 300° C (DUNLOP and STIRLING, 1977; BIQUAND, 1979; KENT, 1985; KENT and MILLER, 1987). This fact agrees with the theoretical unblocking temperatures of VRM of a magnetite bearing sample calculated from the relaxation time versus blocking temperature relations to be about 150–200° C (PULLAIAH *et al.*, 1975) or surprisingly high, of 350–400° C (WALTON, 1980; MIDDLETON and SCHMIDT, 1982) depending on the models. When rocks are heated at upgraded temperature, they acquire a thermo-viscous remanent magnetization which has much higher unblocking temperatures (PULLAIAH *et al.*, 1975). In this case, however, we can usually find a trace of heating (KENT, 1985).

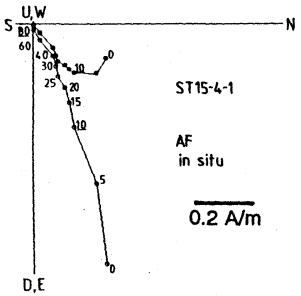
In view of the stable NRMs for both of the AF and thermal demagnetizations and small tilting angle at most sites (usually less than 20°), the directions obtained in this study giving a PEF direction before tilting correction should not be rejected. To support this conclusion, we compared decay curves of AF demagnetization among NRM, ARM, and IRM of some samples. The ARM was induced by a maximum AF field of 80 mT with DC field of 0.05 mT. IRM was induced by putting a sample in a DC field of 5 mT for 1 minute. The maximum DC field of 5 mT was adopted so that the intensity of IRM nearly equals to that of NRM. The AF demagnetization procedure was started after leaving the sample in a non-magnetic space for the following 3 minutes. This procedure was included to stabilize the IRM and to avoid measuring rapidly decreasing components. Figure 4 shows examples of this experiment. The decay curves are normalized by the initial intensity of each remanence and the results of two or three specimens are shown in each diagram. The Zijderveld diagram for in situ coordinates is shown for one example of NRM. NRMs of ST06 are probably of VRM origin and retain no primary components. Samples of ST02 have a dominant VRM component of a direction close to the PEF

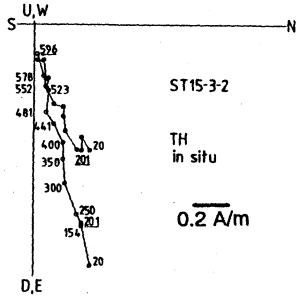

U,W
S
N
D,E
AF
corrected
0.2 mA/m
ST07-2-1

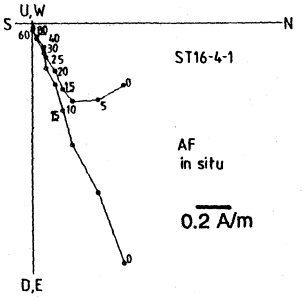

0.05 mA/m
U,W
S
N
D,E
TH (rejected)
in situ
ST07-3-2

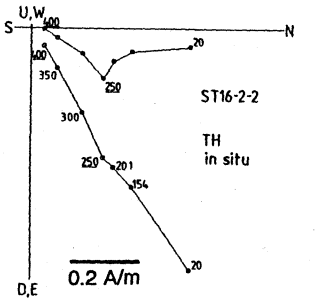

U,W
ST02-8-2
TH
in situ
S
N
50 mA/m
D,E

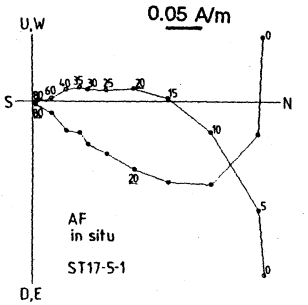

U,W
ST02-7-2
TH
in situ
50 mA/m
S
N
D,E

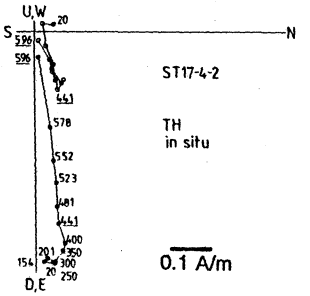

U,W
S
N
ST15-4-1
AF
in situ
0.2 A/m
D,E

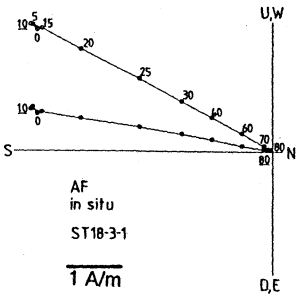

U,W
S
N
ST15-3-2
TH
in situ
0.2 A/m
D,E


U,W
S
N
ST16-4-1
AF
in situ
0.2 A/m
D,E


U,W
S
N
ST16-2-2
TH
in situ
0.2 A/m
D,E


0.05 A/m
U,W
S
N
AF
in situ
ST17-5-1
D,E


U,W
S
N
ST17-4-2
TH
in situ
0.1 A/m
D,E


U,W
S
N
AF
in situ
ST18-3-1
1 A/m
D,E

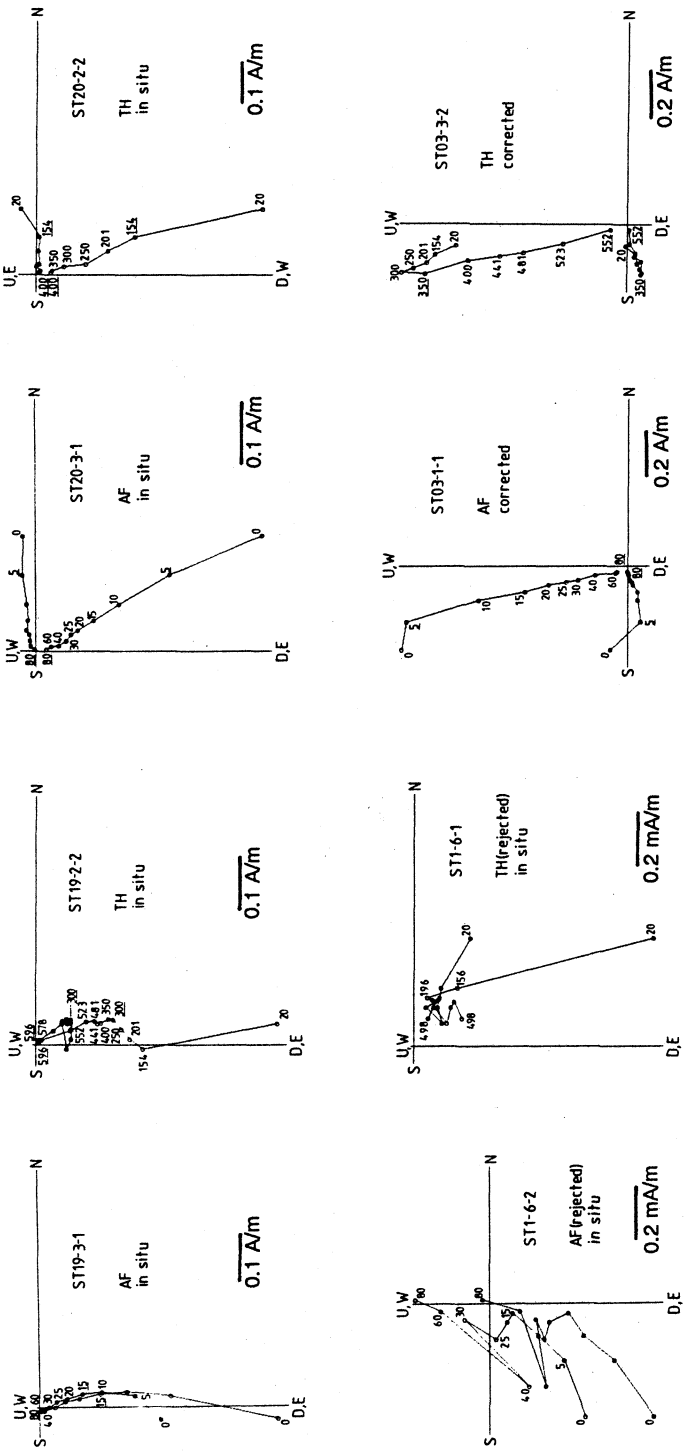
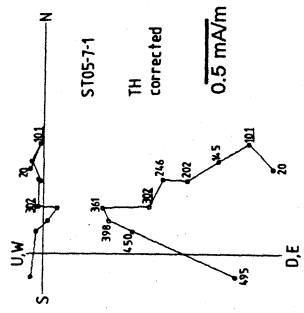
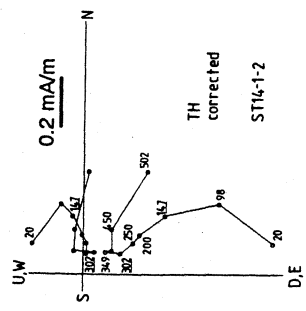
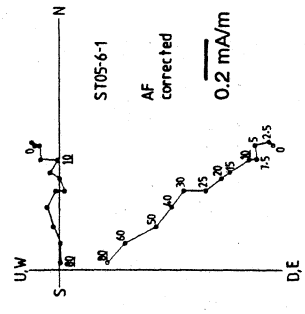
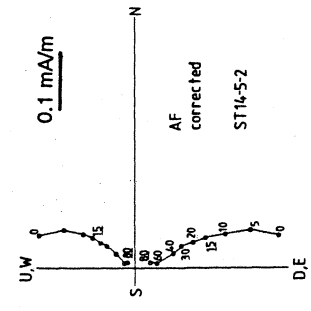


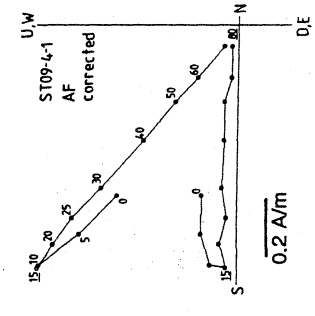
Fig. 3. Examples of progressive demagnetization of NRM by alternating field (AF) and thermal (TH) demagnetizations for all sites. Usually one Zijderveld diagram is shown for each of AF and thermal demagnetizations for each site. Closed and open circles show horizontal and vertical projections, respectively. Whether bedding correction was made or not is shown in each diagram, and scale bar of remanence intensity is also shown. Rejected diagrams because of unstable remanence are shown for unsuccessful sites.

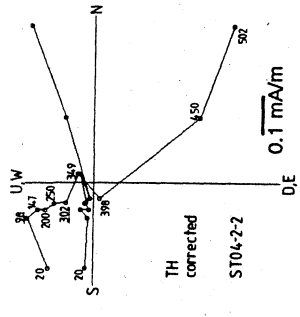

ST05-7-1
TH
corrected
0.5 mA/m
N
S
U,W
D,E

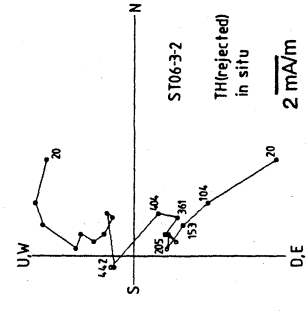

0.2 mA/m
TH
corrected
ST14-1-2
N
S
U,W
D,E

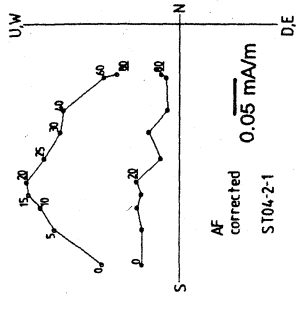

ST05-6-1
AF
corrected
0.2 mA/m
N
S
U,W
D,E

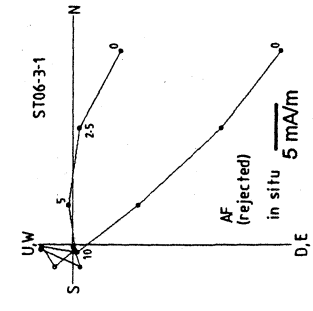

0.1 mA/m
AF
corrected
ST14-5-2
N
S
U,W
D,E

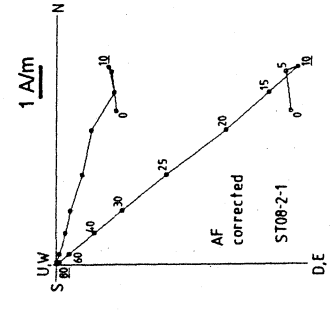

ST09-4-1
AF
corrected
0.2 A/m
N
S
U,W
D,E


0.1 mA/m
TH
corrected
ST04-2-2
N
S
U,W
D,E


ST06-3-2
TH(rejected)
in situ
2 mA/m
N
S
U,W
D,E


0.05 mA/m
AF
corrected
ST04-2-1
N
S
U,W
D,E


ST06-3-1
AF
(rejected)
in situ
5 mA/m
N
S
U,W
D,E


1 A/m
AF
corrected
ST08-2-1
N
S
U,W
D,E

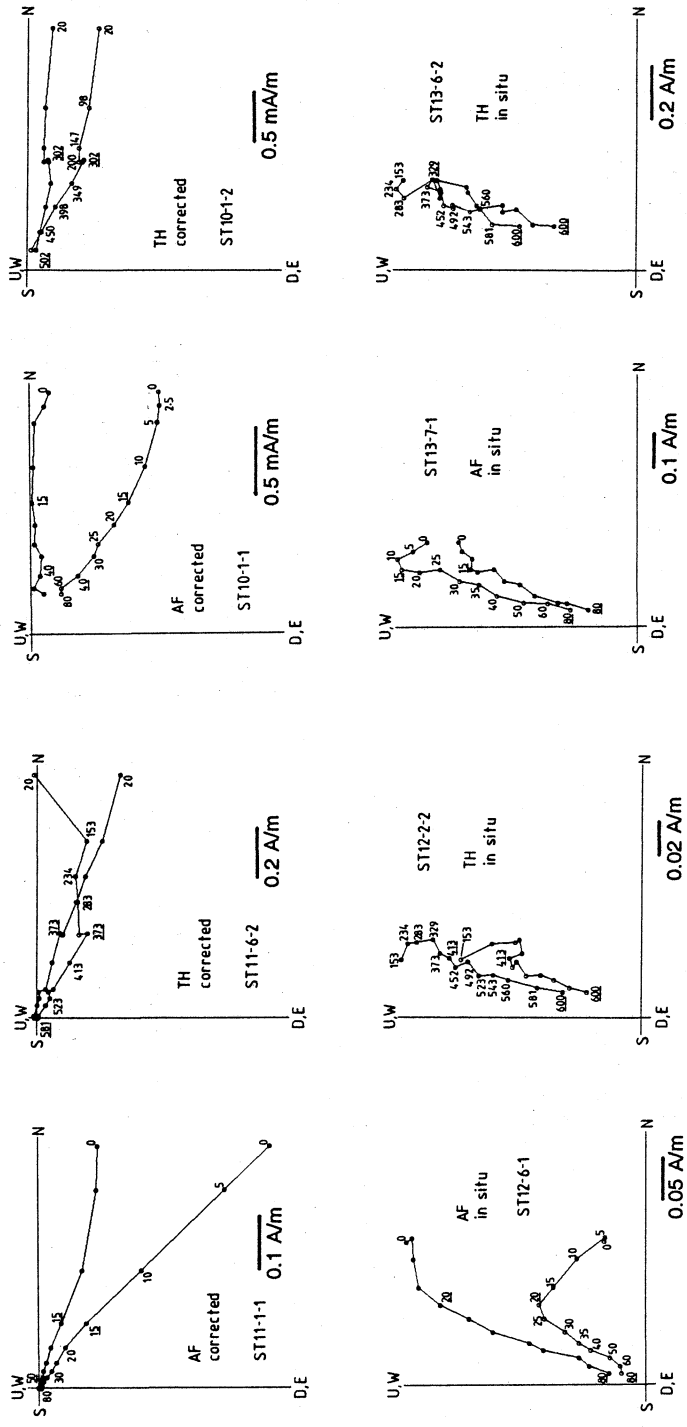


Fig. 3. (continued).

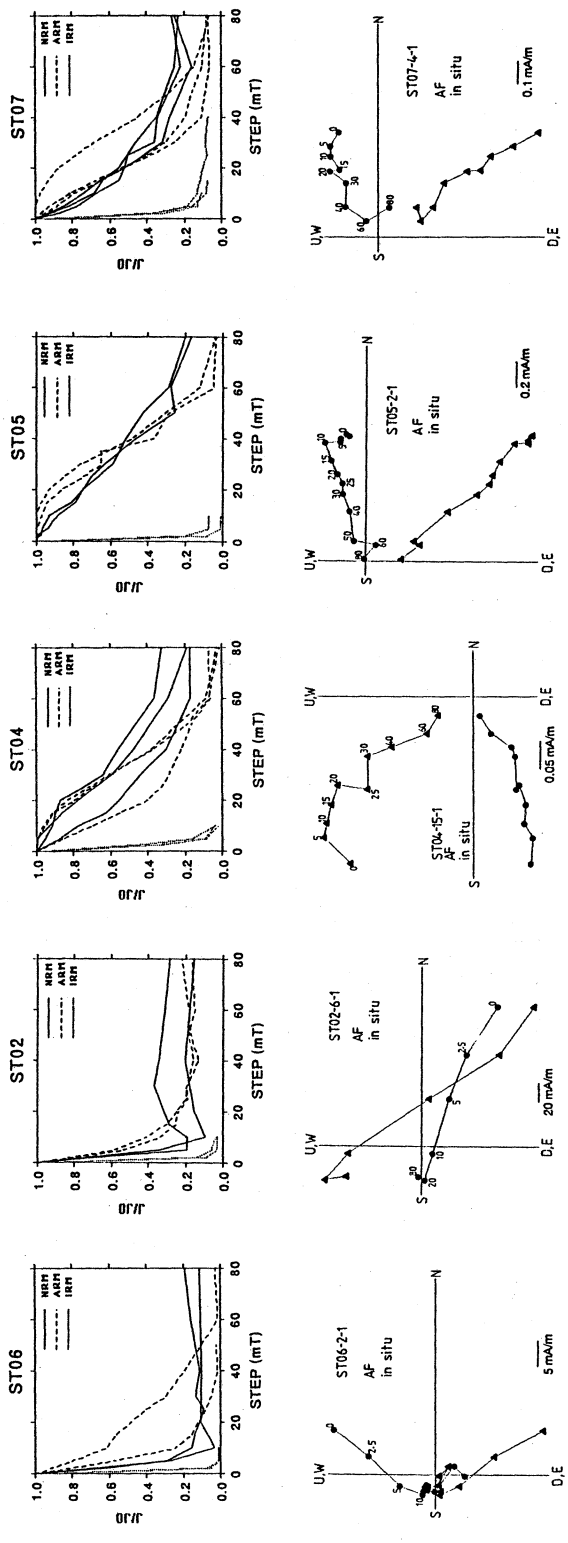


Fig. 4. Comparisons of the decay curves in AF demagnetization among NRM, ARM and IRM. This experiment was made to support the adoption of the in situ direction which is close to the present earth's field (PEF). ARM was induced by maximum peak AF field of 80 mT under DC field of 0.05 mT. IRM was induced by DC field of 5 mT. The Zijderveld diagrams are shown for before bedding correction where circles and triangles are horizontal and vertical projections, respectively. The samples shown are; ST06 (hard shale) which is considered entirely VRM origin, ST02 (hornblende andesite) which has a reversed primary component superposed by a large VRM, ST04 (hard shale) of reversed polarity which itself verifies the non-VRM origin of NRM, and ST05 (hard shale) and ST07 (siltstone) which have in situ directions close to PEF.

direction. The sample is hornblende andesite and the reversed original remanence was revealed by thermal demagnetization. A very quick decay of NRM in the lower steps under 20 mT is similar to that of the IRM in both cases, of ST06 and ST02. ST04 is a hard shale of reversed polarity which itself proves to be of a non-VRM origin. Both NRM and ARM are fairy stable to AF demagnetization and the decay curves are almost overlapped. Two diagrams, of ST05 and ST07, show the result of this test for the hard shale and siltstone samples respectively, of suspicious normal polarity and whose in situ mean site direction contains a PEF direction within its α_{95}. Results of the comparison among the decay curves are very similar to those of the reversed polarity samples verifying its non VRM origin. Although this test is crude because the decay curves of ARM and IRM would depend on various parameters, we think such sort of effort will be necessary. Indeed, recent works by FULLER *et al.* (1988) and HART and FULLER (1988) are the first trials to discriminate between the original and secondary remanences without depending on their paleodirection. They compared the AF demagnetization curves of NRM and saturation IRM (SIRM) on the NRM:SIRM plot of log-log scale. They found out that the larger intensity ratio of NRM and SIRM of lavas with concave down curve is associated with the primary magnetization.

3.3 Paleomagnetic directions

The site mean paleomagnetic directions are summarized in Table 1. Eighteen results out of twenty sites were adopted based on the criteria discussed above. Results of thermal demagnetization were unsuccessful for all of the five siltstone specimens from ST07 of the Yobetsu Formation. We nevertheless adopted the site mean direction determined only by AF demagnetization considering the young age of Pliocene and half solidified material of the siltstone which is often susceptible to chemical change of minerals by heat treatment. As clearly seen in Figs. 5(a) and (b), paleomagnetic directions from volcanic rocks are less convergent than those of sedimentary rocks. This might be caused by the less accurate tilting correction for volcanic rocks and their spot-reading nature of paleomagnetic direction.

There are clearly three anomalous directions in the data from volcanic rocks. To find out the outlier points objectively, HIROOKA *et al.* (1986) proposed an empirical method based on the change of $\alpha_{95}/\log k$ (k is a Fisher's precision parameter; FISHER, 1953) with the number of directions; firstly calculate $\alpha_{95}/\log k$ with the total number, next re-calculate $\alpha_{95}/\log k$ after rejecting the farthest point, and continues this procedure until the number of the points becomes three. In their study, the marked change of $\alpha_{95}/\log k$ was recognized at the stage that all outliers were rejected. In this study, we examined the outlier problem following MCFADDEN (1982) based on statistics. The total mean direction was calculated for 15 sites after removing these three anomalous directions and converting the reversed data to normal. The 95% confidence limit (γ_{95}) on the angle between an outlier and the mean is calculated as 62°. The deviation angles of the outliers (70°, 56°, 61°) are almost close to this value although only one is larger than γ_{95}. If we take a 90% confidence limit, all points exceed or equal γ_{90} of 56°. MCFADDEN (1982) pointed out that his method is not to be a rejection criterion but should be used to decide whether we should search for a possible cause of the error. We think there are some reasons for rejecting these three. The almost horizontal direction (ST17) might reflect an excursion of the geomagnetic field. The reason of the anomalous direction of ST12 and ST13 of the oldest age is unknown. It is difficult to deduce any meaningful paleomagnetic

Table 1. Paleomagnetic site mean directions from the Shakotan Peninsula.

Site	Rock Type (Age) (Ma)	*N*	Demag AF	Demag TH	*I*	*D*	α_{95}	VGP °N	VGP °E	Bedding Attitude	Correction
				Yobetsu Formation (Pliocene)							
ST07	siltstone	6	6/7	0/5	51.7	3.6	14.9	78.6	304.7	N30W15E	tilted
				Toyohama Formation (Late Miocene)							
ST02	andesite (9.2±0.6)*	5	0/8	5/5	−63.0	197.1	8.7	77.7	219.8		in situ
ST15	andesite dike	8	7/8	1/1	55.0	64.8	5.2	40.5	216.1	N30W90[1)]	in situ
ST16	andesite dike	9	8/8	1/1	54.9	51.0	7.4	50.4	223.8	N32W68N[1)]	in situ
ST17	andesite dike	8	8/9	0/1	−2.3	38.4	7.7	33.9	272.0	N15W90[1)]	in situ
ST18	andesite dike	8	8/9	—	−31.7	182.2	3.7	64.0	315.7	N10W85E[1)]	in situ
ST19	andesite dike	9	8/9	1/1	59.3	58.2	6.9	47.3	213.7	N04W72E[1)]	in situ
ST20	andesite dike	10	8/9	2/2	63.6	353.6	11.7	85.0	78.0	N10W85E[1)]	in situ
ST01	siltstone	—	0/8	0/5	(unstable remanence)						
Formation Mean		7 sites			58.0	30.2	17.7	66.5	227.8	(α_{95}=22.7)	
				Furuugawa Formation (Middle Miocene)							
ST03	dolerite (11.7±1.3)*	8	7/7	1/1	−78.8	174.8	3.8	64.8	135.9	N60E08S	tilted
ST04	hard shale	7	5/8	2/2	−59.9	179.9	12.7	87.4	321.7	N54E15E	tilted
ST05	hard shale	4	2/7	2/5	53.8	7.7	13.3	79.3	284.2	N08E22E	tilted
ST06	hard shale	—	0/5	0/5	(unstable remanence)						
Formation Mean		3 sites			64.2	2.7	20.5	85.2	158.9	(α_{95}=28.9)	
				Upper Furubira Formation (Early Miocene)							
ST14	mudstone	9	7/8	2/2	67.1	319.4	11.2	61.6	78.4	N10W23W	tilted
ST08	basalt	8	8/8	—	48.9	21.7	4.1	68.1	261.2	N45W20E[2)]	tilted
ST09	basalt (15.1±1.5)*	6	6/6	—	−39.4	185.5	5.8	68.6	306.5	N45W20E[2)]	tilted
ST11	basalt	7	6/8	1/1	37.5	21.8	7.5	61.3	274.4	N45W20E[2)]	tilted
Formation Mean		4 sites			50.1	8.4	24.3	78.5	293.4	(α_{95}=31.1)	
				Lower Furubira Formation (Early Miocene)							
ST10	mudstone	7	5/8	2/2	33.3	3.9	7.6	64.8	311.9	N78E32N	tilted
				Kayanuma Formation (Early Miocene)							
ST12	rhyolite (35.3±2.6)**	9	8/8	1/1	−31.6	295.9	7.3	6.0	20.4	no attitude data[3)]	
ST13	rhyolite (35.3±2.6)**	8	7/8	1/1	−54.0	302.4	10.1	3.7	184.8	no attitude data[3)]	
Total mean (ST17, 12, 13 are rejected)		15 sites			55.6	14.7	9.5	77.9	249.5	(α_{95}=12.1)	

Notes; [1)]attitude of dike relative to the present horizontal plane. [2)]general bedding plane of host sedimentary rocks. [3)]in situ paleodirections are presented. *data by K-Ar dating. **data by Fission Track method.

N: number of samples, Demag: Demagnetization method, AF: AF demagnetization, TH: Thermal demagnetization, n1/n2 in the Demag columns means that there was n1 successful specimens out of n2 total specimens subjected to the progressive demagnetization.

direction because both sites have no tilting data due to their massive structure.

The site mean paleomagnetic directions for all sites after rejecting the three anomalous sites are shown in Fig. 5(c). Almost north-south antipodally distributed paleodirections are clearly seen, and the total mean paleomagnetic direction becomes (I=55.6°, D=14.7°, α_{95}=9.5°) after converting the reversed data to normal. There is no special grouping of the paleodirections according to the formation (formation means are

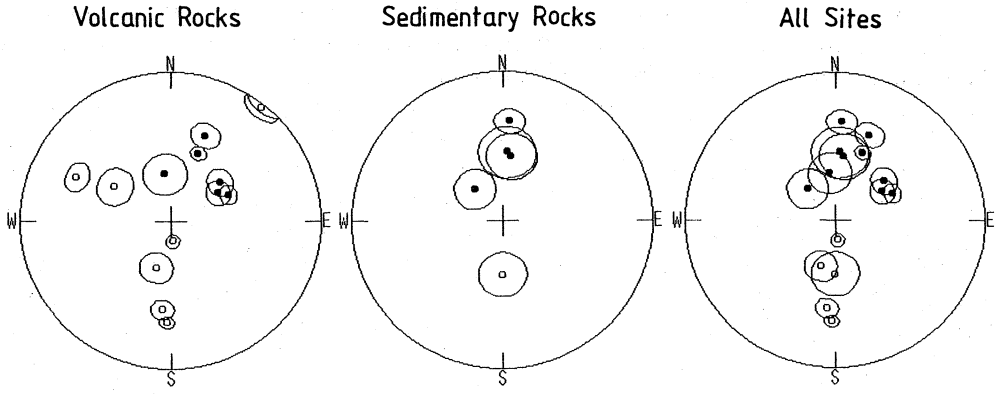


Fig. 5. Site mean directions plotted on Lambert equal-area projection. (a) All of 13 sites from volcanic rocks were successful to reveal stable remanences but 3 sites show anomalous directions. (b) 5 sites out of 7 sedimentary rocks were successful, but other 2 sites were rejected because of unstable remanences. (c) Compiled site mean directions after rejecting 3 anomalous directions from volcanic rocks.

included in Table 1). The exceptions are the results from three dikes we will discuss later. We conclude one single paleomagnetic pole (77.9°N, 249.5°E, $\alpha_{95}=12.1°$) which was calculated from virtual geomagnetic poles (VGPs) of all fifteen sites. The age of this pole covers from uppermost Early Miocene (around 15 Ma) to Pliocene, although it is difficult to confine the exact age of the oldest limit.

4. Discussion

The paleomagnetic direction obtained in this study is a little eastward as known from the positive mean declination of 14.7°. However, we do not think the deflection from north-south direction is significant, because the α_{95} of the corresponding mean pole contains the geographical pole. We think, instead, that the mean direction was biased by the three directions from dike swarms (ST15, ST16, and ST19) which show almost identical directions of large eastward declination. We suspect these three dikes intruded during a short volcanic episode at almost the same time although any geological evidences were not found. It is one of the problems in the paleomagnetism of volcanic rocks that several directions might not be independent in terms of time to average out geomagnetic secular variation effect. This is caused by collecting multiple units formed within a brief volcanic episode as already reported for basalt extrusions in Hawaii and Australia (MCWILLIAMS *et al.*, 1982; IDNURM, 1985). If the three directions are reduced to one by averaging them ($I=57.0°$, $D=56.5°$, $\alpha_{95}=3.9°$), the total mean direction is re-calculated as ($I=54.2°$, $D=8.7°$, $\alpha_{95}=9.6°$) from the 13 sites. The corresponding paleomagnetic pole is (81.3°N, 273.1°E, $\alpha_{95}=11.5°$). It is noted that the sedimentary rocks which are better to cancel out the geomagnetic secular variation effect give almost north-south paleomagnetic directions (Fig. 5(b)).

The obtained paleomagnetic pole for 5–15 Ma does not indicate any large tectonic movement of the Shakotan Peninsula for this time range. In the eastern part of Hokkaido, it is considered that the collision of Kuril Arc to Tohoku Arc occurred since the Late Miocene (KIMURA and TAMAKI, 1985). As the time range of our pole includes

the proposed period of the collision, the implication of our result to the tectonics of Hokkaido is, that the tectonic process which was so large to build the Hidaka Mountains did not propagate as far as to the Shakotan Peninsula.

The first paleomagnetic study from West Hokkaido was described by KAWAI *et al.* (1971) who studied Quaternary and Neogene rocks and suggested a possible block movement of Hokkaido relative to Northeast Japan based on the easterly deflected paleomagnetic direction. It is known that West Hokkaido is geologically a continuation of Northeast Japan (IIJIMA, 1964; SEGAWA and FURUTA, 1978), and we had better leave out the data of KAWAI *et al.* (1971) because they made no remark on whether any demagnetization procedures were made. Nevertheless it is worthwhile to ascertain the tectonic affinity of West Hokkaido with Northeast Japan by paleomagnetism.

The paleomagnetic pole of this study for 5–15 Ma does not conflict with the integrity of West Hokkaido and Northeast Japan and this is because Northeast Japan had finished the block rotation before 15 Ma. It is much clearer by the result of FUJIWARA and SUGIYAMA (1986) that West Hokkaido and Northeast Japan were in the same block. They studied Miocene to Pliocene volcanic and sedimentary rocks from the Oshima Peninsula, southern part of West Hokkaido, and reported on four paleomagnetic poles covering the time span of 7–24 Ma which age was based on fission track dating (KOSHIMIZU *et al.*, 1986). Poles A (<11.2 Ma), B (11.2–14.5 Ma), C (14.5–17.1 Ma), and D (21.7–24.3 Ma) of FUJIWARA and SUGIYAMA (1986) are shown on an equal area projection in Fig. 6 with that of this study (S, 5–15 Ma), and the details are summarized in Table 2. These clearly indicate that West Hokkaido had gone through counter clockwise rotation and probably ceased at around 15 Ma. Figure 7 shows poles from both, West Hokkaido and Northeast Japan. The closed circles with α_{95} drawn by a solid line and the squares with α_{95} by a dotted line indicate the poles of West Hokkaido and Northeast Japan, respectively. Three younger poles A, B, and S were averaged to a mean pole W for 5–15 Ma (80.1°N, 280.5°E, α_{95}=10.4°). Poles from Northeast Japan were roughly numbered in the ascending order of their ages and included in Table 2. It is noted, at a glance of Fig. 7, that the two pole groups of West Hokkaido and Northeast Japan completely overlap. This must strongly support that West Hokkaido as the

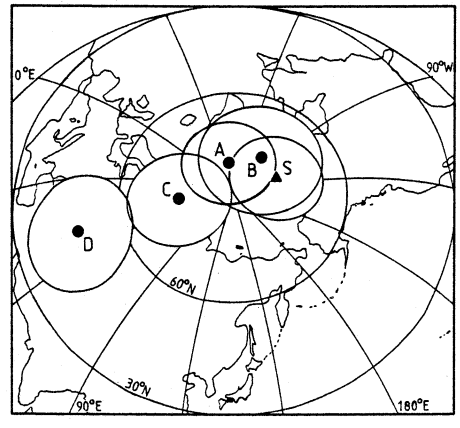


Fig. 6. Lambert equal-area projection of paleomagnetic poles from West Hokkaido by FUJIWARA and SUGIYAMA (1986) (A: 7–11 Ma, B: 11–14.5 Ma, C: 14.5–17 Ma, D: 20–25 Ma) and this study (S: 5–15 Ma).

Table 2. Oligocene-Pliocene paleomagnetic poles from Northeast Japan and West Hokkaido.

ID	Formation	Age (Ma)	Location		Site	Pole Position			Ref.
			°N	°E		°N	°E	α_{95}	
			—West Hokkaido—						
A	Assabu Formation	7–11	42	140.5	11	80.8	322.7	11.7	a
B	Kikonai Formation	11–14.5	42	140.5	5	77.6	280.6	15.3	a
S	Shakotan Peninsula	5–15	43.2	140.5	15	77.9	249.5	12.1	*
W	mean pole of A, B, S	5–15				80.1	280.5	10.4	
C	Kunnui Formation	14.5–17	42	140.5	16	75.8	47.7	13.3	a
D	Fukuyama Formation	20–25	42	140.5	7	48.1	51.4	14.9	a
			—Northeast Japan—						
1	Shimokura dike swarm	8	38.3	140.7	15	75.0	81.0	7.0	b
2	Nishikurosawa and Onagawa Formation	10–17	40	140	7	88.6	220.5	7.0	c
3	Matsushima and Otsuka Formation	15–17	38.3	141.1	6	79.5	296.6	9.3	d
4	Daijima Formation	18–25	40	140	6	47.2	55.8	13.2	c
5	volcanic rocks	20–32	36–40	139–142	17	57.0	56.0	12.8	e
6	Monzen Formation	28–42	40	140	6	59.1	51.7	12.9	c

Notes; Site: number of sites.

References: *) this study, a) FUJIWARA and SUGIYAMA (1986), b) TSUNAKAWA *et al.* (1985), c) TOSHA and HAMANO (1988), d) YAMAZAKI (1989), e) OTOFUJI *et al.* (1985a).

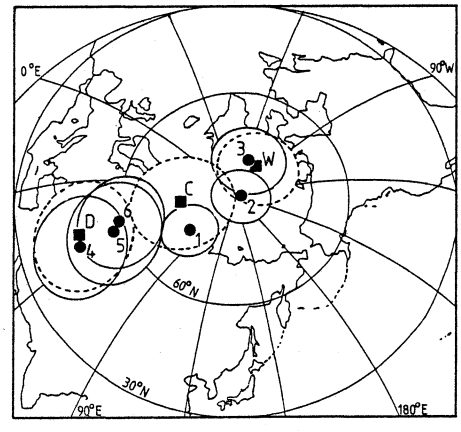


Fig. 7. Comparison of paleomagnetic poles from Northeast Japan (closed circle) and West Hokkaido (square) for about 5–40 Ma. Pole W (5–15 Ma) is a mean pole of A, B, and S for West Hokkaido. Poles from Northeast Japan are roughly numbered in ascending order of age.

northernmost part of Northeast Japan.

Examination of Fig. 7, however, reveals some inconsistency of the data. Pole 1 (8 Ma) from the Shimokura dike swarm (TSUNAKAWA *et al.*, 1985) and pole 3 (16 Ma) from the Matsushima and Otsuka Formations (YAMAZAKI, 1989) suggest much different implications to that of the rotation of Northeast Japan in spite of their close sampling

localities; the former suggests that the rotation continued until 8 Ma or younger while the latter suggests that the rotation had finished by 16 Ma. It is not easy to solve this discrepancy, and we need of more paleomagnetic data from various parts of Northeast Japan.

It is difficult to conclude the timing of rotation of Northeast Japan as precisely as that of Southwest Japan, even though obviously it is around 15 Ma. We suspect, however, that the rotation of Northeast Japan occurred earlier than Southwest Japan. This idea was also suggested by YAMAZAKI (1989) based on his paleomagnetic study of the Matsushima region, and the paleodirection of ST10 (~15 Ma) of this study probably supports this. We think it is not self-evident to suppose the synchronous rotations of Northeast and Southwest Japan associated with the opening of the Japan Sea if the rotation of Northeast Japan had been affected much more by the opening of the Japan Basin than that of Yamato Basin. Actually TAMAKI (1986) suggested some amount of time lag between the openings of the two basins (earlier for Japan Basin) based on the data of topography and heat flow, although a recent geochronological study by KANEOKA *et al.* (1990) did not find any age difference among dredged volcanic rocks from the two basins.

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